Wave 6 — crescent [crescent_smoke_qd2] | Fitness 2.557 (SPG×A×(1+N)) | SPG 3.458 N/g | Est. weight 84.9 g

Detailed plate FEM (physics::fine DKT bending + T6 membrane) with per-element thickness field (taper, grinds, relief, lattice, fuller, flutes, texture). Capacities CERTIFIED = min over 0.25 / 0.125 mm² meshes; robust = min(20 °C at −0.1 mm stock, 100 °C yield)

AESTHETIC FLOOR: PASS
H 1.58 mm · EMD 0.005 · A 0.691

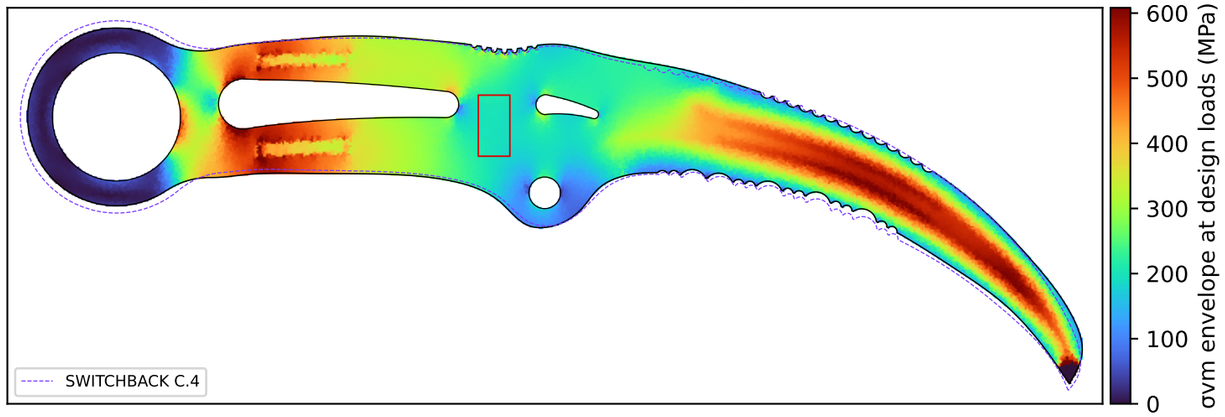
BLADE ≥ 3.375 in on all three: PASS
min 3.450 in

HARD GATES: PASS

FLOORS 180 / 260 / 1500 N (certified): PASS
lat 294 N · ring 16505 N

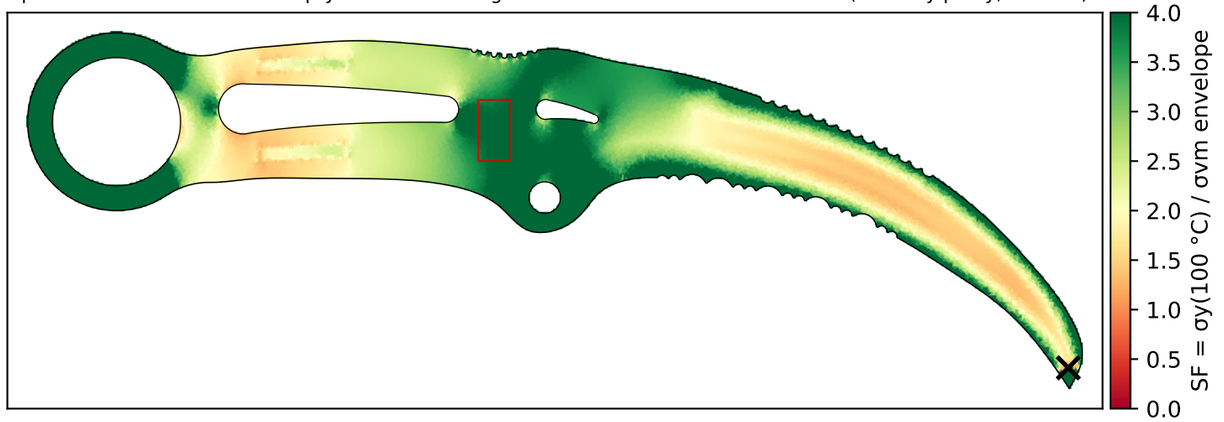
NOVELTY 0.070
(k-NN, aesthetic passers only)

von Mises envelope at design loads: lateral 260 N, pry 260 N, twist 1.26 N·m, ring pull 1500 N (4 mm tip-load zone excluded)

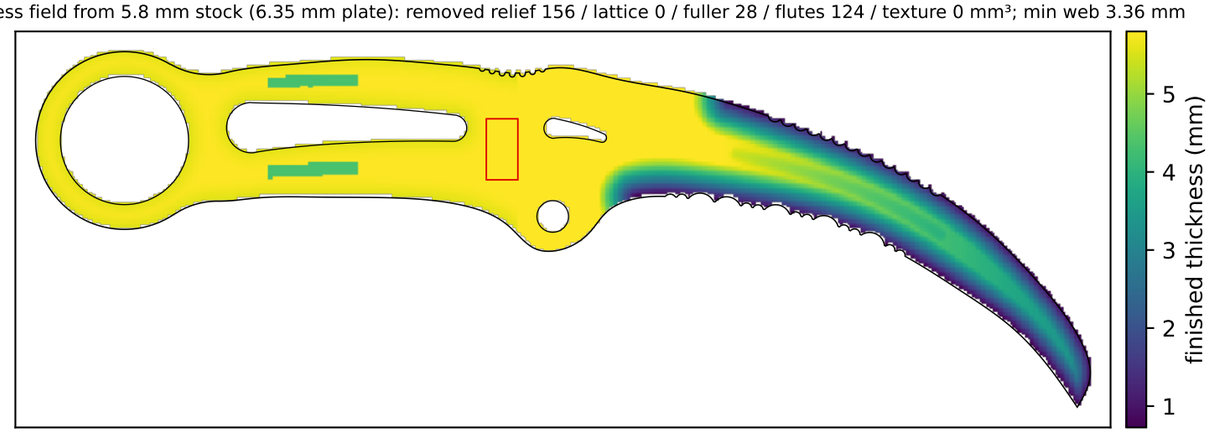


tip→heel	3.811 in (96.8)
horizontal reach	3.450 in (87.6)
edge arc inside / spine	3.989 / 3.943 in
grip / stock / Ti	61.3 mm / 5.8 mm / Ti-6Al-4V
weight (finished / blank)	84.9 g / 101.5 g
lateral 20 °C / robust	322 / 294 N
SF hiker 180 / hunter 260	1.63 / 1.13
ring pull 20 °C / robust (SF)	18097 / 16505 N (11.0)
tip pry 20 °C / twist 20 °C	694 N / 1.25 N·m
SF 100 °C lat / pry / ring / twist*	1.15 / 3.04 / 12.33 / 0.93
governing lateral at	(25, -3) mm
holder box (oriented)	198.9 × 99.0 mm @ -9.0°
aesthetic A / H / EMD	0.691 / 1.58 mm / 0.005
straight run / min blend R	16.0 / 9.1 mm
novelty / SPG	0.070 / 3.458 N/g

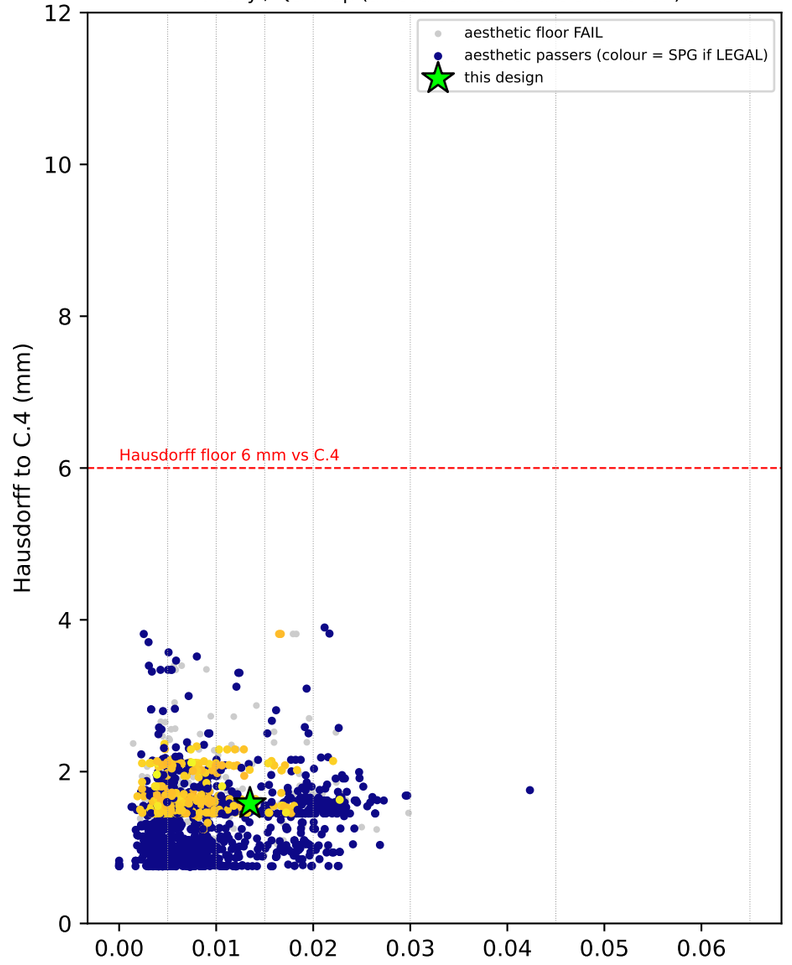
safety factor at 100 °C, envelope of all four cases (min 0.93 at x=188, y=-49 mm)
per case: lateral 260 N 1.15 · pry 260 N 3.04 · ring 1500 N 12.33 · twist 1.26 N·m 0.93 (advisory proxy, no floor)



thickness field from 5.8 mm stock (6.35 mm plate): removed relief 156 / lattice 0 / fuller 28 / flutes 124 / texture 0 mm³; min web 3.36 mm



novelty / QD map (all evaluations of the smoke run)



source out/crescent_smoke | author Jepson Taylor | generator ops/wave_figures/make_crescent_plate.py | utility / hiking karambit only